ECE 3317

Homework #7

1) A plane wave in free space has an electric field given by

$$\underline{E} = \left(4\underline{\hat{x}} - 4\underline{\hat{y}} - \sqrt{0.8}\underline{\hat{z}}\right)e^{-j\left(k_x x + k_y y + k_z z\right)}, \text{ where } k_x = 0.4k_0 \text{ and } k_y = 0.2k_0 .$$

Determine k_z and the $\underline{k}$ vector, and verify that the electric field is perpendicular to the $\underline{k}$ vector. Then determine the magnetic field vector in the phasor domain from Maxwell's equations, and verify that it is also perpendicular to the $\underline{k}$ vector.

$k_z = \sqrt{0.8}k_0 = 0.8944k_0$

$\underline{k} = \hat{x}(0.4k_0) + \hat{y}(0.2k_0) + \hat{z}(0.8944k_0)$

$\underline{H} = \dfrac{k_0}{\omega\mu}e^{-j\underline{k}\cdot\underline{r}}\left(\hat{x}3.399 + \hat{y}3.935 - \hat{z}2.4\right)$

2) Calculate the critical angle for a plane wave going from:

(a) Water $\left(\varepsilon_r = 81, \mu_r = 1\right)$ to air.

(b) Silicon $\left(\varepsilon_r = 11.68,\ \mu_r = 1\right)$ to air

(a) 6.38°

(b) 17.01°

3) White light is composed of the entire visible spectrum. The index of refraction n for most materials is a weak function of wavelength λ, often described over a limited frequency region by Cauchy's equation, $n = A + B/\lambda^2$. A beam of white light is incident at 35° on a piece of glass with $A = 1.4$ and $B = 6\times10^{-15}\ \text{m}^2$. What are the transmitted angles for colors violet (400 [nm]), blue (450 [nm]), green (550 [nm]), yellow (600 [nm]), orange (650 [nm]), and red (700 [nm])? This separation of colors is the dispersive effect of a prism.

violet	blue	green	yellow	orange	red
23.52°	23.65°	23.83°	23.88°	23.93°	23.96°

4) A microwave signal at 2.0 GHz from a cell-phone base station bounces off of a building, which is modeled as a lossless nonmagnetic material having a relative permittivity of $\varepsilon_r = 4$. Assume that the angle of incidence is 45^{o}. What is the percent power reflected if the wave is a TM_z polarized wave? What is the percent power reflected if it is a TE_z polarized wave? What is the percent power reflected if it has equal powers in both polarizations?

TM_z: 4.15%, TE_z: 20.38%.
Equal powers: 12.26%